Sensitivity Modeling of 2DEG AlGaN/GaN Hall-Effect Sensors under Nuclear Radiation

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I. INTRODUCTION

HALL-EFFECT sensors have a wide array of applications, including velocity and position sensing for the automotive industry, and current sensing in power electronics. The low-cost, easy integration, and linearity over a wide range of magnetic fields, make Hall-effect devices a great choice for magnetic field sensing [1, 2, 27].

Electronic components are typically made of silicon (Si) due to its low cost, ease of fabrication, and compatibility with integrated circuits. Nevertheless, there is a growing need for Hall-effect sensors that can operate under extreme conditions, such as in outer space and nuclear applications. An example consists in in-situ magnetic field monitoring of fusion experiments inside a quasi-helically symmetric magnetic field stellarator, where in-vessel radiation is present due to the plasma [58]. Therefore, previous Hall-effect device studies [1, 2] have proposed to replace Si with wide bandgap materials, such as gallium nitride (GaN) for high temperature operation. In opposite to silicon-based components with a breakdown beyond 200°C [28, 29, 30], GaN-based Hall-effect sensors have shown reliable operation up to 12 hours at 576°C [2].

Next to extreme temperatures, devices employed in the previously cited environments are also exposed to both particle and electromagnetic radiation, such as electrons, protons, neutrons, and gamma rays [20, 25, 12]. GaN with its more tightly bound atoms and higher lattice displacement threshold energy (low phonon loss and high threshold for electron-hole pair generation upon ionizing radiation [40]) is more radiation-hard to displacement damage and total ionizing dose effects than Si and gallium arsenide (GaAs) [12, 23, 24, 25, 30-34]. Three main radiation effects are relevant and devices should withstand them: single-event effects (SEEs), total ionizing dose (TID), and displacement damage (DD). SEEs are due to single energetic particles. TID and DD are cumulative effects related to the ionizing dose and the particle fluence, respectively [39, 46].

For the previously cited reasons, in this study GaN Hall-effect sensors are studied. GaN heterostructures have a two-dimensional electron gas (2DEG) that is formed when a nanometers-thick layer of unintentionally doped aluminum gallium nitride (AlGaN) is deposited on an underlying GaN layer [1]. The 2DEG, created from differences in the polarization fields of the III-nitride layers [35, 36], has a high

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Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org electron mobility [37], which enables high sensitivity devices and high linearity on bias voltage [38].

In this work, the combination of high doses of neutron and gamma irradiation, typically dealt with in nuclear reactors [20], will be modeled using technology computer-aided design (TCAD) software in AlGaN/GaN Hall-effect plates. Gamma rays (photons) primarily produce TID, while neutrons, emitted from fusion and fission, create DD (atoms will be displaced in crystal lattice, and structural defects will be induced) [9]. Gamma radiation is important in hardness assurance testing of devices devoted to space equipment [15, 16], while a combination of neutron and gamma rays is well suited for simulation of military environments and the nuclear power industry [17, 18].

II. Simulation Setups

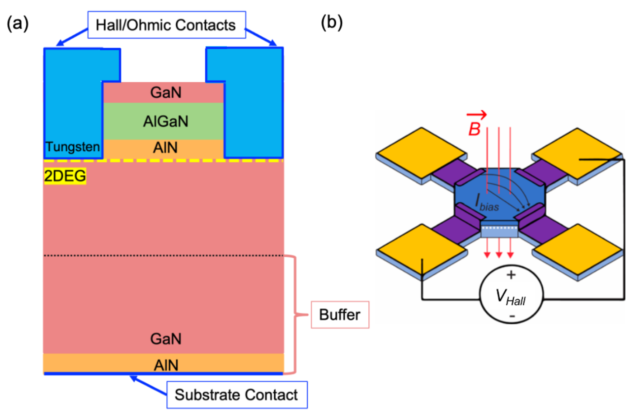


Fig. 1. (a) Cross-sectional schematic of the simplified $Al_{0.25}Ga_{0.75}N$/GaN Hall Device. (b) Operation principal of Hal-effect device with applied radiation.

A. Structure Definition and Model Calibration

Octagonal Hall-effect plates with equal sides and a diameter of 100 μm were simulated using the technology computer-aided design (TCAD) software *Sentaurus Synopsys*.

It has been chosen to model devices previously fabricated in the lab [1, 2] for calibration. Those consist in AlGaN/GaN Hall-effect sensors giving optimal voltage sensitivity (S_v). The AlGaN/GaN stack consists of a 2 nm GaN cap, a 30 nm $Al_{0.25}Ga_{0.75}N$ barrier layer, a 1 nm AlN spacer and a 1.2 μm GaN layer. In the model, a simplified 1.5 μm GaN/AlN buffer layer and tungsten ohmic and hall contacts are used. This stack is represented in Fig. 1. The simulated, room temperature and non-radiated, sheet resistance is 328.3 $\Omega/\blacksquare$, the sheet electron density is 8.38×10^{12} cm^{-2}, and the electron mobility is 1806.85 $cm^2/V{\cdot}s$ which corresponds to what has been obtained in [1, 2]. This results in a current and voltage sensitivity of respectively $S_i = 54.67$ V/A/T and $S_v = 93.74$ V/V/T.

B. Principal of Operation

To determine the sensitivity of Hall-effect sensors, the Hall voltage (V_H) is measured perpendicular to the applied current (I) and external magnetic field (B). It is defined as

$$V_H = \frac{IBr_nG_H}{qn_s} \tag{1}$$

where r_n is the scattering factor of the material ($r_n \approx 1.1$ for GaN [3]), G_H is the geometry-dependent shape factor that accounts for the short-circuiting effects of having finite contacts [4, 5, 6] (for equal lengths octagonal-shaped devices, $G_H = 0.667$ [1]), q is the electronic charge and n_s is the electron sheet density.

The sensitivity of a Hall-effect device with respect to supply current (S_i) is inversely proportional to n_s,

$$S_i = \frac{V_H}{IB} = \frac{r_n}{qn_s}G_H. \tag{2}$$

In addition, the sensitivity with respect to the supply voltage (S_v) is proportional to the electron mobility (μ_H),

$$S_v = \frac{V_H}{V_sB} = \frac{r_nG_H}{Rqn_s} = \mu_H r_n \frac{G_H}{\left(\frac{L}{W}\right)_{eff}}, \tag{3}$$

where V_s is the supply voltage, R is the device resistance, and $(L/W)_{eff}$ is the effective number of squares, defined as the ratio of the internal resistance to the sheet resistance [7].

C. Device Radiation

Both neutron and gamma radiations are studied in this work. Neutron irradiation primarily causes DD, introducing acceptor-like traps (point defects) in GaN devices [9, 10]. The primary traps type is V_{Ga}-related, due to its relatively low formation energy [11]. Defects induced in the AlGaN and GaN layers by neutron radiation degrade the carrier mobility through Coulomb interactions and degrade the carrier concentration n_s through charge compensation and carrier removal [8, 9, 12].

The reduction in n_s can be explained through a reduction of the initial donor dopants (n_0) within the channel region as

$$n_{eff} = n_0 - n_{traps}, \tag{4}$$

where n_{traps} corresponds to the concentration of acceptor traps introduced by neutron irradiation, and n_{eff} the effective doping after irradiation. n_{traps} can be defined through both the fluence (ϕ) and the carrier removal rate (η_e) with

$$n_{traps} = \phi\eta_e, \tag{5}$$

where it is assumed that displacements collisions between impinging protons and lattice atoms leads to the formation of defects that decreases the proportion of conduction electrons in proportion to the fluence [13, 19].

The increased scattering from those traps reduces the electron mobility. This can be modeled using the bulk trap Coulomb scattering (BTCS) mobility model (μ_C) [14] which includes the dependency on n_{traps}, the influence of the screening by mobile charge carriers, and the weakening of Coulomb scattering with increasing temperature (T) as

$$\mu_C = \frac{\mu_1\left(\frac{T}{300K}\right)^k \left\{1+\left[\frac{n_{eff}}{n_{trans}\left(\frac{n_{traps}}{n_{ref}}\right)^{\eta_1}}\right]^{\nu_1}\right\}}{\left(\frac{n_{traps}}{n_{ref}}\right)^{\eta_2}}, \qquad (6)$$

where μ_1, k, n_{trans}, n_{ref}, η_1, η_2 and ν_1 are fitting parameters. The total electron mobility (μ_{tot}) is given by Matthiessen's rule, considering the non-radiated mobility (μ_0), as

$$\frac{1}{\mu_{tot}} = \frac{1}{\mu_0} + \frac{1}{\mu_C}. \qquad (7)$$

Gamma irradiation, on the other end, mainly creates TID. The ^{60}Co γ photons (stable isotope), when they interact with weakly bound electrons when some electric field is applied, create an ionization effect through the generation of high-energy Compton electrons (loss of energy to release electrons). Without an external electric field, electron-hole pairs generated by irradiation through the ionization effect recombine rapidly [39, 41].

This electric field-dependent electron-hole pairs generation (G_r) can be modeled as

$$G_r = g_0 D \cdot \left(\frac{F+E_0}{F+E_1}\right)^m, \qquad (8)$$

where D is the dose rate, g_0 is the generation rate of electron-hole pairs, F is the electric field, and E_0, E_1 and m are fitting constants [14].

Nuclear applications show can high-dose gamma-ray irradiation with ^{60}Co-γ rays above 1 MeV. While the energy is lower compared to the neutron irradiation (meaning lower non-ionizing energy loss, NIEL), when the photon has kinetic energy higher than the displacement threshold energy of GaN, high mean energy Compton electrons can induce DD, introducing defects [42, 46]. Under these conditions, the incoming secondary electron can displace lattice atoms, creating scattering effect, which in turn can produce Frenkel pairs and defect clusters which can migrate, recombine, or form complexes, changing the strain state within the material [39, 41, 47]. In opposite to neutron irradiation, ^{60}Co-γ is mainly (more than twice as much for Compton electrons of energy 600 keV) related to V_N defects instead of V_{Ga} as nitrogen atoms have a lower displacement energy. For neutron irradiation, the V_N traps are negligible as V_{Ga} acts as a triple acceptor. Those V_N vacancies create nitrogen interstitials (Frenkel pair) which act as deep acceptor [46]. This similarly generates carrier removal and mobility degradation [19, 45]. Therefore, the effective dopants and mobility follow (4), (5) and (6) respectively, with ϕ, the photon fluence in this case.

In this study, both TID and DD are studied, with TID called the "in-situ" experiment and DD the "ex-situ" one.

D. Temperature Influence

Néglige les local temperature increase mais pourrait faire une analyse Générale d'augmentation de temperature pour montrer que les augmentations d'énergies locales ont un effet d'annealing (donc augmentation de la mobilité + je sais pas effet sur ns)

III. Results and Discussion

The GaN Hall devices studied in this work have different material layers the radiations must go through before getting to the 2DEG. Therefore, it is important to study the attenuation of the rays, studying the variation between the transmitted dose after travelling through a thickness x ($I(x)$) with regards to the initial radiation intensity (I_0) given by times x?? (formula)

$$I(x) = I_0 \exp\left(-\frac{\mu}{x}\right), \qquad (9)$$

where μ is the atomic cross section. The ^{69,71}Ga stable isotopes, at 14 MeV (typical energy for neutron radiations [8, 12]), as well as at 1 MeV energy (for gamma radiation [42, 46]) gives an attenuation smaller than 0.01% Need to be done for Ga, Al, N (different websites). Therefore, the attenuation has been neglected in the device.

A. Ex-Situ Operation (DD)

DD, caused by neutron radiation, has been simulated through the insertion of acceptor-like traps. The concentration is defined in (5), with energy levels of $E_C - 0.45$ eV and $E_C - 1.0$ eV in the AlGaN and GaN layers respectively [8, 9, 10].

Fluences in typical nuclear operations range in the order of 10^{14} to 5×10^{16} cm^{-2} at room temperature [9, 10, 12, 26] and GaN carrier removal rate for n-doped GaN around 10^{15} cm^{-3} of $\eta_e \approx 1$ cm^{-1} [10].

To simulate the DD caused by neutron radiation, the acceptor-like trap concentration is defined through (5) with energy levels of $E_C - 0.45$ eV and $E_C - 1.0$ eV in the AlGaN and GaN layers respectively [8, 9, 10].

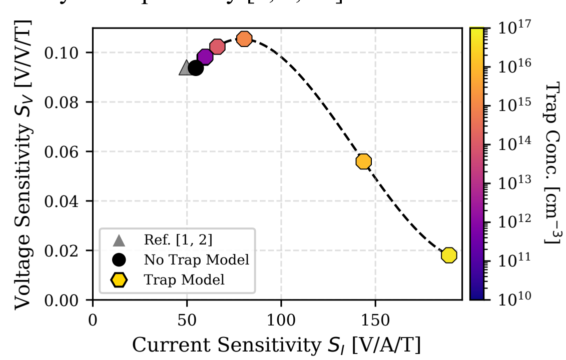


Fig. 2. Voltage-scaled and current-scaled magnetic sensitivity for various neutron irradiation induced trap concentrations. The black (no trap model) and grey ([1, 2]) comparison serves for benchmarking of the model. The colored octagon-shaped points show the obtained values for trap concentrations going from 10^{10} to 5×10^{16} cm^{-3}, corresponding to similar fluence values, supposing $\eta_e = 1$ cm^{-1}. Add different values of nref.

Fig. 2 shows the impact of increasing neutron-induced trap concentration on sensitivity. Firstly, one can observe a great model calibration of the model without traps with previous AlGaN/GaN Hall-effect devices studies [1, 2].

Next, the current sensitivity increases with trap concentration, due to a decrease in the 2DEG sheet density n_s

in (2). This is due to the fact that adding acceptor traps in the channel, reduces the effective concentration as described in (4).

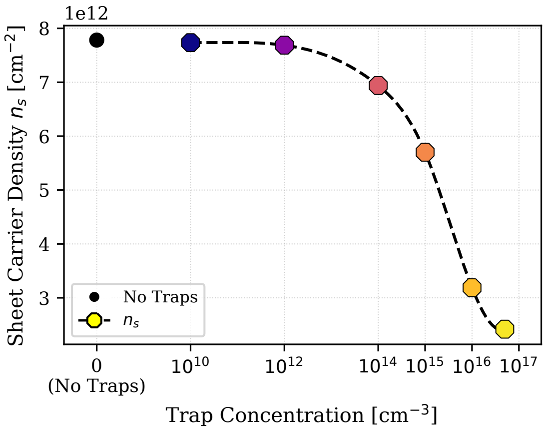


Fig. 3. Sheet carrier density n_s vs. trap concentration.

Voltage sensitivity, on the other end, shows a slight increase up to a trap concentration around 10^{15} cm^{-3} before decreasing at higher concentrations. At low trap concentrations (lower than the intrinsic doping of the GaN layer), the scattering effect is low and thus induces a negligeable decrease in electron mobility, as represented by μ_e in Fig. 4, and modeled through a sufficiently high value of n_{ref} in (6). However, those acceptor-like traps neutralize background impurities (donor dopants) that are originally present, reducing current leakage, creating a slight improvement in voltage sensitivity source? + lien avec autres papers. This can be seen in μ_H (Fig. 4), the Hall mobility, extracted from (3). While both Hall and electron mobilities, as defined in this work, should be the same in theory, thanks to the r_n and G_H parameters in (3), as those parameters have been kept constant, the influence of the trap neutralization is observed in μ_H.

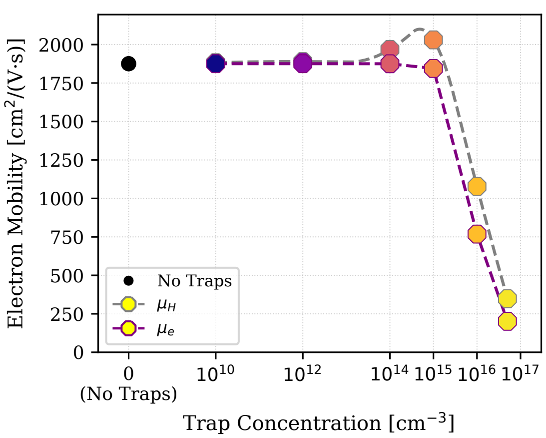


Fig. 4. Hall (μ_H) and 2DEG electron (μ_e) mobilities vs. trap concentration. One observes a decrease in μ_e with increasing trap concentration linked to increase scattering effects, while μ_H shows some neutralization around 10^{15} cm^{-3}.

The decrease in mobility with device depth, is represented in Fig. 5. One can observe that the decrease is "faster" for higher trap concentrations, directly linked to a decrease in 2DEG width (also explaining the decrease in n_s) previously, as n_{eff} decreases.

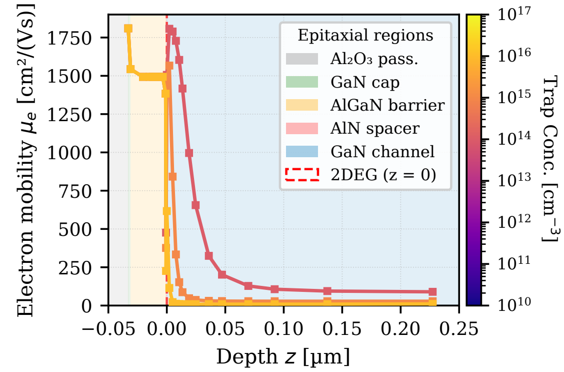


Fig. 5. Electron mobility μ_e in the different layers of the device (close to the 2DEG), for three trap concentrations (10^{14}, 10^{15} and 10^{16} cm^{-3}).

DD with gamma radiation happens at high doses, mentioned around 10 kGy(Si) up to 1000 kGy(Si) ([41] shows no induced traps below 6 kGy(Si), while [55, 56] show a device degradation due to traps at 700 and 1000 Gy(Si)), for particle accelerators or nuclear reactors [45, 49, 53, 54] at energy levels of approximately 1 MeV [39, 42, 48], giving deep trap levels at energies around $E_C - 1$ eV, which corresponds to the same level as for nuclear radiation. Therefore, the previous study is also valid for DD in gamma radiation. Pas correct – donor traps

Gamma radiation is defined by absorbed dose, instead of particle fluence like neutron radiation [48, 49]. Therefore, the trap concentration is given by the dose and the production rate. The silicon (Si) equivalent absorbed dose can be linked with the fluence (photon flux) in (5), through 1 kGy(Si) = 1.9 × 10^{14} cm^{-2} using 2.7 × 10^{-3} m^2/kg as the mass energy absorption coefficient of Si for 1.25-MeV photons [50]. Also, instead of talking about carrier removal rate, the production rate, which means trap generation rate, is used in (5) and is close to $\eta_e \approx 10^{-3}$ cm^{-1} for the deep level traps [48, 49, 51]. For extreme doses of 1000 kGy(Si), this gives an acceptor trap concentration in the order of $n_{traps} = 10^{14}$ cm^{-3} which is at least an order magnitude lower than for nuclear radiation happening simultaneously [48]. This means that the DD impact of the gamma radiation is negligible compared to nuclear radiation.

B. In-Situ Operation (TID)

TID is mainly observed when an electric field is present, and thus, within the nuclear reactor for instance. Dose rates within RMC's SLOWPOKE-2 reactor are up to 111 krad/hr for 50% power [52]. TID without DD is usually observed with doses up to 600 krad(Si) (or 6kGy(Si)) or 100 rad/s [41]. Current defense system requirements for TID are 300 krad(Si) [39].

Expliquer avec le photocurrent + ionizing current

2 cases : total ionizing dose (TID) and photocurrent:
1) TID:

A) Need traps (in the top layers):

- Compton electrons induced from y-radiation create electron-hole pairs, thus changing occupancy of traps.[48]
- Under the action of 60Co γ photon, the ionization effect is mainly through the generation of high-energy Compton electrons, and then through various mechanisms to produce a large number of secondary

electron-hole pairs. The subsequent trapping/detrapping process will affect the dynamic transport behaviors of carriers and change the occupancy of traps, leading to the deterioration of dynamic threshold voltage (VTH) and the decrease of carrier concentration and mobility [57]

- Since the Hall-effect device has metal contacts, and no oxides like in typical silicon devices, it does not suffer from oxide damage effects and is thus tolerant to total ionizing radiation dose effects, which can explain the reduced influence [43, 44].

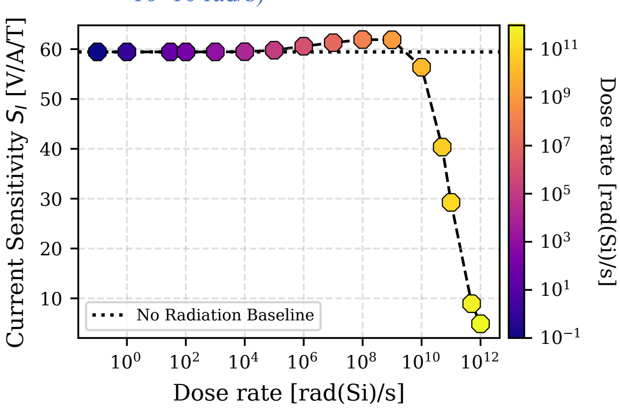
B) Only in top layers (2DEG rapid recombination)

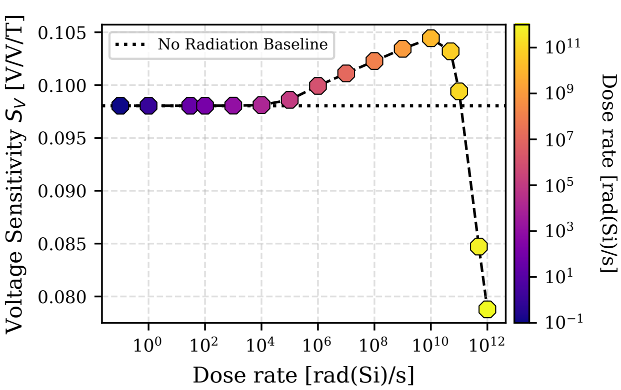
- Without an external electric field, electron-hole pairs generated by irradiation through the ionization effect recombine rapidly [57]
- The negative shift of the threshold voltage mainly comes from the hole trapping in the barrier layer or the interface between the barrier layer and the channel layer [46]

→ impact due to trap occupation and not prompt photocurrent

2) Photocurrent – without any traps

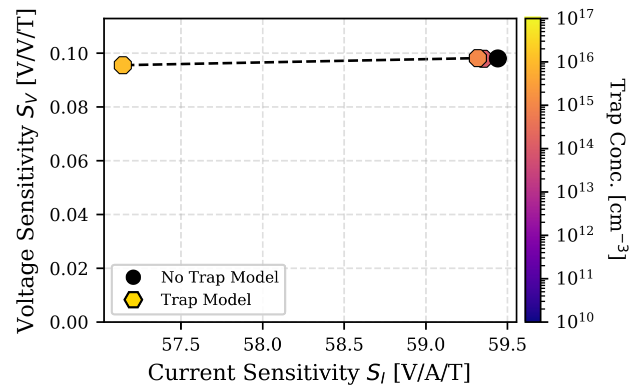
- Fast recombination in 2DEG → no impact
- In top layers, quasi-fermi level splitting → creates a potential difference, but only impacts Si and Sv, if a parallel path, sufficiently high to induce some changes and induce this extra current (around 10^10 rad/s)





3) DD

- Knocks out VN which consists in donor traps
- 0.6 MeV Compton electrons, generated by the interaction of the material with 60Co gamma-photons at average energy of 1.25 MeV, are likely to create donor-type nitrogen vacancy-related defects in III-N layers. These types of defects have been reported after low-energy proton, electron, and gamma irradiation. [45]



- Comparison with Fd-SOI → dielectric (comparison with paper)
- Put TID and DD together for gamma radiation to be able to compare with literature
- Not doing temperature study
- Cite Tim in the paper
- Neutron radiation, no TID because neutron is not creating hole-pairs due to its neutral charge

Sensitivity study (=> concentration and mobility : can recover from sentaurus) + resistance study

Add annealing (maybe temperature effects)

IV. Conclusion

References

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